

Ardalan Kahak

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PROFESSIONAL SUMMARY

Mechanical Engineering Ph.D. candidate at Virginia Tech specializing in electromechanical system design, soft robotics, and prototype development. Hands-on experience spanning programmable acoustic holograms, inflatable Kirigami interfaces, soft piezoelectric reservoir-computing patches, and haptic interfaces. Skilled in embedded electronics, sensor integration, fabrication, and control systems, with additional industry experience in large-scale automation and manufacturing. Targeting advanced R&D, soft robotics, and wearable electronics roles where deep fabrication expertise and cross-disciplinary systems thinking drive product innovation.

EDUCATION

Virginia Tech <i>Doctor of Philosophy (Ph.D.) in Mechanical Engineering.</i>	Blacksburg, VA <i>December 2022 – Present</i>
Sharif University of Technology <i>Bachelor of Science in Mechanical Engineering.</i>	Tehran, Iran <i>September 2017 – July 2022</i>

EXPERIENCE

Manufacturing Engineer / Intern <i>Durali System Design and Automation (DSDA)</i>	July 2020 – September 2022 <i>Tehran, Iran</i>
- Led design, manufacturing, and installation of large-scale theater stage automation and safety systems. - Designed the conceptual architecture of a high-speed automated mask manufacturing line to address surging demand for 3-ply masks during the COVID-19 pandemic. - Executed CAD modeling, prototype fabrication, system assembly, and manufacturing documentation (BOMs) for a compression-chain lifting mechanism. - Coordinated with external vendors and led post-installation troubleshooting to improve system performance and reliability.	
Founder <i>Poultry Breeding Project</i>	July 2019 – September 2019 <i>Shahriyar, Tehran, Iran</i>
Founded and managed a small-scale poultry breeding operation incorporating automated temperature and humidity control systems, optimized feeding logistics, and experimental environmental monitoring.	

PROJECTS

Ph.D. Thesis: Programmable Kirigami Interfaces <i>Researcher</i>	September 2023 – Present
- Fabricated prototypes and mechanical assemblies with integrated sensors using layer-by-layer fabrication and geometry-driven principles for applications in acoustics, haptics, soft robotics, and physical reservoir computing. - Designed, built, and validated a low-powered underwater physical reservoir computing sensor by integrating embedded sensors, piezoelectric materials, and DAQ systems alongside mechanical design optimization. - Developed an inflatable compliant Kirigami interface with embedded sensors for closed-loop feedback control of exoskeleton comfort, using a single pneumatic input for actuation and force regulation.	
Bachelor's Thesis: Reinforcement Learning of a Planar Micro-swimmer <i>Researcher</i>	September 2021 – July 2022
Developed and trained a reinforcement learning framework in Python for a triangular micro-swimmer model operating in viscous microscale fluid environments for potential biomedical navigation applications.	
Machine Element Design Projects <i>Student Engineer</i>	
Completed seven mechanical design projects involving real-world engineering applications, including the design of a heavy-duty dump truck rear differential, gearbox, and motor selection for a mine elevator.	

SKILLS

Fabrication and Prototyping: Soft material fabrication, Silicone rubber casting, Laser cutting, Embedded electronics prototyping, Flexible electronics, Digital fabrication, Transfer printing
Electromechanical: PCB design, Feedback and motion control, PID control
Mechanical Design: SOLIDWORKS, COMSOL, Abaqus, Product Development, Design for Manufacturing (DFM)
Instrumentation: Data Acquisition (DAQ), Sensors, Signal conditioning
Programming: MATLAB, Python, C, C++, Arduino
Hardware and Experimental Tools: Universal testing machines, Oscilloscopes, Function generators, Laser/plotter cutter machines, FDM/SLA/DMP printers

TEACHING

- Teaching Assistant, Strength of Materials | Sharif University of Technology, Dept. of Mechanical Engineering | February 2021 – March 2022
- Technical and Vocational Instructor | Allamah Helli 5, National Organization for Development of Exceptional Talents | January 2018 – September 2021: Designed and supervised hands-on engineering projects including robotic grippers and water rocket launcher systems for high school students. Organized and managed a mathematics and puzzle-based competition with weekly evaluations, team scoring, and final tournament coordination.

LEADERSHIP AND ACTIVITIES

- Debating and public speaking on technology, education, and interdisciplinary engineering topics.
- Athletics:** wrestling, swimming, resistance training, and soccer.